

Federated Multiagent Reinforcement Learning for Resource Allocation in NR-V2X Mode 2

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Abstract—The Third Generation Partnership Project (3GPP) introduced cellular vehicle-to-everything (C-V2X) for vehicular communications. In the standard, C-V2X Mode 4 is defined for the distributed resource selection. Subsequently, in 3GPP Release 16, NR-V2X is introduced with Mode 1 and Mode 2 for vehicular communications. Likewise C-V2X Mode 4, NR-V2X Mode 2 is used for decentralized resource scheduling. The vehicles select the resources based on their local observations by utilizing the semi-persistent scheduling (SPS). Since, the vehicles select the resources based on the local observation, sensing nature of SPS is challenged by the hidden node problem that lead to resource conflict. To resolve the contention, 3GPP also introduced the physical sidelink feedback channel (PSFCH) to assist the distributive resource scheduling based on the receiver feedback. However, this incurred a signaling overhead. In this work, federated learning is exploited for distributive training via offline method and distributive multiagent-based resource scheduling is performed following the principles of NR-V2X Mode 2. Distributed training favors the model accuracy by accommodating the varying affect of the environment due to the high mobile dynamics. Simulation is conducted by integrating SUMO in conjunction with 3GPP NR-V2X standard. Performance results demonstrate a substantial improvement compared to other deep learning methods, where centralized training and random resource selection procedures are employed. This research marks a significant stride toward efficient and conflict-resilient resource allocation in vehicular communications.

Index Terms—Distributive training, federated learning, NR-V2X Mode 2, semi-persistent scheduling (SPS).

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I. INTRODUCTION

VEHICLE-TO-EVERYTHING (V2X) communication, an integral part of future intelligent transportation systems (ITS), is pivotal for enabling various safety, traffic management, and autonomous driving applications. V2X supports real-time exchange of information between vehicles (V2V), infrastructure (V2I), pedestrians, and networks, creating a connected transportation environment. These applications help reduce road accidents and optimize traffic flow, playing a key role in the evolution of smart cities and the Internet of Vehicles (IoV). Within the realm of V2X technologies, Cellular V2X (C-V2X) has gained prominence due to its ability to provide reliable communication in both in-coverage and out-of-coverage scenarios, facilitating safety and infotainment services in real-world driving environments.

C-V2X was introduced in Release 14 by the Third Generation Partnership Project (3GPP) to support basic safety applications across diverse scenarios, such as in-coverage and out-of-coverage communication [1]. In C-V2X Mode 3, resource allocation is centrally managed by a base station (eNB or gNB) in in-coverage scenarios, whereas C-V2X Mode 4 employs distributed resource allocation in out-of-coverage regions, allowing vehicles to independently choose transmission resources through a semi-persistent scheduling (SPS) approach. Despite its advantages, C-V2X Mode 4 faces challenges, particularly in the medium access control (MAC) sublayer, where inefficiencies arise when managing aperiodic messages of varying sizes [2].

To address these limitations and support more advanced vehicular services, 3GPP introduced 5G new radio (NR)-V2X in Release 16 [3]. NR-V2X introduced improvements, including sidelink communication, which allows direct communication between vehicles (referred to as terminal nodes or user equipments (UEs) without routing through the core network. However, it is important to note that NR-V2X sidelink communications are designed to complement, rather than replace, C-V2X sidelink communications [4], [5].

Within the NR-V2X MAC sublayer, 3GPP introduced two modes of operation: 1) NR-V2X Mode 1 and 2) NR-V2X Mode 2. NR-V2X Mode 1 mirrors C-V2X Mode 3, wherein resources are scheduled by the base station. On the other hand, NR-V2X Mode 2, a distributed resource allocation mode, builds upon C-V2X Mode 4 by incorporating a mandatory re-evaluation mechanism [6]. Despite these advancements, challenges in NR-V2X Mode 2 remain. For example, while

research studies [7], [8], [9], [10] have evaluated the effectiveness of the SPS scheme, difficulties persist in efficiently disseminating aperiodic traffic. A long-term sensing window may become obsolete for high-mobility vehicle UEs (VUEs), while reducing the sensing window may increase the probability of resource unavailability in dense vehicular networks. Additionally, variable message sizes can lead to resource reselection, causing underutilization of previously reserved resources. Another major challenge is the hidden node problem in SPS, which can result in contention during the resource allocation process [11].

In Release 17 of NR-V2X, a feedback channel was introduced to improve transmission reliability. However, resource contention issues still persist, as the feedback mechanism introduces additional signaling overhead [12]. This highlights the need for more efficient solutions to manage resource contention and optimize resource allocation in high-density vehicular environments.

This manuscript is organized as follows. Section II outlines the motivation and contributions. Section III reviews related work. The system model and proposed scheme are presented in Sections IV and V, respectively. Section VI covers the performance evaluation. Section VII provides a complexity analysis of the proposed scheme and compares it with both centralized and other decentralized methods. Finally, the conclusion is drawn in Section VIII.

II. MOTIVATION

In the rapidly evolving landscape of ITS, efficient and reliable resource allocation remains a critical challenge, particularly in decentralized systems like NR-V2X Mode 2 [3]. In this Mode, vehicles select communication resources independently based on local observations, which increases the likelihood of resource contention due to spectrum scarcity and decentralized decision-making [13]. This contention degrades network performance, reducing communication reliability and efficiency.

To address this, federated learning offers a scalable and efficient solution by enabling roadside units (RSUs) to train local models based on vehicular state information within their coverage areas. Instead of transmitting raw data to a central controller, RSUs send model updates (e.g., learned weights or gradients) for aggregation into a global model, which is then disseminated back to vehicles. This decentralized approach reduces computational and communication complexity, allowing parallel learning at RSUs and minimizing the central controller's load [14], [15]. By continuously updating the global model, the system adapts to dynamic vehicular network conditions, ensuring accurate resource allocation policies even as traffic patterns and interference levels change.

Compared to game theory, which relies on equilibrium models and frequent agent interactions, federated learning excels in reducing signaling overhead and computational demands. Game theory-based methods often struggle to adapt to highly dynamic vehicular environments due to their reliance on fixed assumptions and iterative processes. In contrast, federated learning's ability to aggregate local insights into a global

model ensures scalability and adaptability, making it better suited for NR-V2X Mode 2 applications [16].

Federated learning also reduces signaling overhead compared to traditional feedback mechanisms, such as those used in SPS with physical sidelink feedback channel (PSFCH). Vehicles rely on periodically updated global models for autonomous resource allocation, eliminating the need for constant signaling and improving network efficiency. While privacy is a known advantage of federated learning, the primary focus here is on enhancing efficiency. This decentralized framework aligns with NR-V2X Mode 2's operational principles, offering scalable, adaptive, and efficient resource allocation in high-mobility vehicular environments.

A. Contribution

Federated learning has been applied in various V2X scenarios, including resource allocation in V2V and V2I communications, mode selection, and computational offloading for vehicular edge computing. However, much of this work, such as in [17] and [18], focuses on maximizing V2I performance or optimizing offloading, without addressing the specific resource scheduling and contention challenges in NR-V2X Mode 2. Some studies [19], [20], [21] explore federated learning for vehicle selection or energy efficiency, but these approaches overlook decentralized resource scheduling in NR-V2X Mode 2. While works like [22], [23] examine joint communication and resource management, they do not tackle the distinct challenges posed by NR-V2X Mode 2. In contrast, this article introduces a novel federated learning framework designed to complement the NR-V2X Mode 2 standard for decentralized resource selection, specifically addressing the challenges of resource contention and spectrum scarcity.

To the best of the authors' knowledge, this is the first work to exploit federated learning in NR-V2X Mode 2, where vehicles select communication resources in a decentralized manner from preconfigured resource pools allocated via system synchronization messages exchanged with the gNB over the system synchronization block (SSB). To enhance the resource selection process, we propose a federated learning approach that leverages distributive training at RSUs and aggregates these local models to develop a global policy, which vehicles then use for autonomous resource selection. Our approach ensures that vehicles select resources in a distributive fashion, following the NR-V2X Mode 2 principles, without relying on a central controller (gNB) during the decision-making process. The reward function is specifically designed to optimize network performance, while Shannon's theorem is incorporated to improve the system's capacity. Additionally, this framework significantly improves exploration efficiency, allowing vehicles to more effectively explore and select optimal resources in a decentralized manner. The contributions of this work are as follows.

- 1) This work pioneers the use of federated learning to enhance decentralized resource selection in NR-V2X Mode 2, reducing resource contention and improving overall efficiency. The framework performs distributive training at RSUs, capturing local vehicular state

information, which is aggregated into a global model. This global model is disseminated to vehicles, enabling them to make autonomous, distributive resource selection decisions without constant reliance on a central gNB.

- 2) A reward function is introduced to maximize network performance, ensuring efficient and optimal resource utilization across the network. Shannon's theorem is considered to maximize the system's capacity, improving data rates and managing spectrum resources more effectively in dense vehicular environments.
- 3) The proposed approach enhances exploration efficiency by distributive training allowing vehicles to more effectively explore and select the best resources within the preconfigured resource pools, minimizing collisions and optimizing resource usage.
- 4) The proposed federated learning framework is highly scalable, allowing the system to accommodate larger vehicular networks, while dynamically adapting to changing traffic conditions and mobility patterns, ensuring reliable performance in high-mobility scenarios.

III. RELATED WORK

In response to the challenges posed by aperiodic traffic of varying sizes in V2X communications, [24] presents a congestion control strategy combining listen-before-talk (LBT) and decentralized congestion control (DCC) for autonomous resource allocation under high channel load. This approach enhances resource management but does not address decentralized resource scheduling in NR-V2X Mode 2, where resource contention remains critical. Similarly, [25] introduces a DCC algorithm using deep reinforcement learning (DRL) to optimize VUE communication, achieving improvements in packet delivery ratio (PDR) and service level time (SLT). However, it lacks focus on the decentralized resource contention issues.

To further explore resource-sharing challenges in C-V2X, [1] investigates Quality-of-Experience (QoE) considerations and connectivity modeling, proposing an uncertainty-oriented connectivity model. This model addresses the challenges of mobility, dynamic environments, and service provision, contributing to the understanding of resource-sharing dynamics and QoE optimization in C-V2X. Although insightful, this work does not focus on the decentralized resource scheduling challenges in NR-V2X Mode 2.

Moreover, MAC layer enhancements in 5G NR V2X aim to address potential packet collisions and inefficiencies seen in long term evolution (LTE) V2X MAC under variable aperiodic traffic conditions. However, challenges persist, particularly when handling diverse traffic patterns not fully considered in previous evaluations [7]. Addressing these diverse traffic patterns in NR-V2X remains critical for efficient resource management in dynamic environments.

In the context of aperiodic message transmissions, vehicles often need to reselect resources when the message size exceeds the initially reserved resources, especially in time-sensitive scenarios [3]. In such cases, resource reselection poses a

significant challenge in maintaining reliable communication. Additionally, conflicts may arise during resource scheduling within SPS when vehicles schedule without detecting each other due to hidden nodes. To address this issue, [12] introduces a receiver grant-based resource allocation method, leveraging the PSFCH in NR-V2X, enhancing transmission reliability and reducing collision ratios.

Abubakar et al. [26] focused on evaluating the limitations of 5G V2V physical sidelink shared channel (PSSCH) capacity, which is important for cooperative awareness message (CAM) exchange in 5G-V2X. Using M/G/1 queuing theory and Newton Raphson method, the study assesses the impact of CAM payload sizes and vehicle densities on PSSCH capacity. The findings highlight the importance of managing CAM payload sizes to maintain PSSCH messages below 1500 bytes, ensuring optimal capacity demands below 1 Gb/s. This study reveals the complex interplay between CAM payload sizes, PSSCH message sizes, and their implications for overall system capacities.

Additionally, [27] proposes a strategic approach to mitigate mutual interference in coexisting IEEE 802.11p and LTE-V2X sidelink transmissions. This methodology employs an IEEE 802.11p preamble, which is compatible with upcoming standards, to optimize sidelink communication amidst spectrum scarcity challenges.

Federated learning has been applied to various aspects of V2X communications. For example, [17] applies federated learning to maximize the sum rate in V2V and V2I communications but does not consider the decentralized resource scheduling challenges specific to NR-V2X Mode 2. Similarly, [18] employs federated learning for Mode selection in 5G-V2X, yet it lacks a focus on NR-V2X resource scheduling. Federated learning has also been used for computational offloading and resource management in 6G-V2X networks [19], [20], though these works do not address resource contention in NR-V2X Mode 2.

Further, studies, such as [28] and [29] propose methods for resource management and QoE optimization in Device-to-Device (D2D) communication and connected vehicles, but they do not target the unique decentralized scheduling demands of NR-V2X Mode 2. While [30] addresses latency challenges in C-V2X using a two-tiered vehicular fog computing framework, it does not tackle resource contention in a decentralized manner.

To enhance SPS in NR-V2X, [31] proposes Enhanced SPS (eSPS) using reinforcement learning to adapt resource reservations based on aperiodic message generation patterns. Meanwhile, [32] introduces a framework using decentralized multiagent markov decision processes and actor-critic schedulers to dynamically adjust resource allocation in C-V2X networks. However, neither study addresses decentralized resource selection under the NR-V2X Mode 2 standard in high-mobility environments.

Authors in [37] and [38] adopt a distributed resource allocation approach employing multiagent DRL, where each vehicle acts as a learning unit. They train a centralized model shared by the vehicles, enabling them to learn from one another's experiences. However, this approach does not fully

consider varying V2X traffic patterns in different directions. Numerous studies have attempted to solve resource allocation problems [39], but resolving resource contention remains a persistent challenge in NR-V2X.

Recent studies have also explored federated learning for V2X resource allocation. For instance, [33] minimizes interference between cellular users and VUEs using federated learning, and [22] incorporates vehicle mobility into federated learning for resource management. However, these works do not fully address the decentralized resource scheduling required in NR-V2X Mode 2, where vehicles select resources independently without a central controller.

Other recent works include [34], which explores federated learning for sparse code multiple access (SCMA) in V2X, focusing on optimizing codeword design and complexity reduction. However, this work does not address resource contention in NR-V2X Mode 2. Similarly, [35] presents a federated reinforcement learning-based approach for delay-sensitive IoV services, using a multiagent deep Q network (DQN) to optimize spectrum-energy efficiency. While it addresses spectrum and power allocation, it does not consider decentralized resource selection. Finally, [36] compares classical optimization methods with machine learning techniques like multiagent reinforcement learning for V2X resource allocation. It does not consider the NR-V2X standard and federated learning.

In NR-V2X Mode 2, decentralized resource allocation presents several critical challenges, particularly in dynamic and high-density vehicular networks. Resource contention arises when vehicles independently select resources based on local observations, often leading to overlapping resource usage in the absence of global coordination. Additionally, the highly dynamic nature of vehicular environments, characterized by fluctuating traffic densities, rapid mobility, and varying interference levels, requires resource allocation mechanisms to adapt quickly to changing conditions. Scalability is another significant challenge, as traditional centralized methods struggle to manage the computational and signaling overhead in large-scale networks. Moreover, high signaling requirements in conventional approaches increase communication delays and hinder real-time decision-making, particularly in dense networks.

To the best of the authors' knowledge, this article is the first to apply federated learning for distributed training in NR-V2X Mode 2, addressing both mobility and decentralized resource scheduling challenges. By leveraging distributive training at RSUs, this work enhances exploration efficiency and reduces resource contention, optimizing network performance in a fully decentralized manner without relying on a centralized gNB, making it ideal for dynamic vehicular environments. Vehicles execute resource allocation decisions autonomously based on pretrained global policies, ensuring scalability regardless of network size. The periodic re-evaluation of local conditions at RSUs ensures the global model adapts dynamically to network changes, while efficient communication through model updates minimizes signaling overhead. This approach significantly improves network reliability and efficiency, offering a robust solution tailored to the unique challenges of NR-V2X Mode 2.

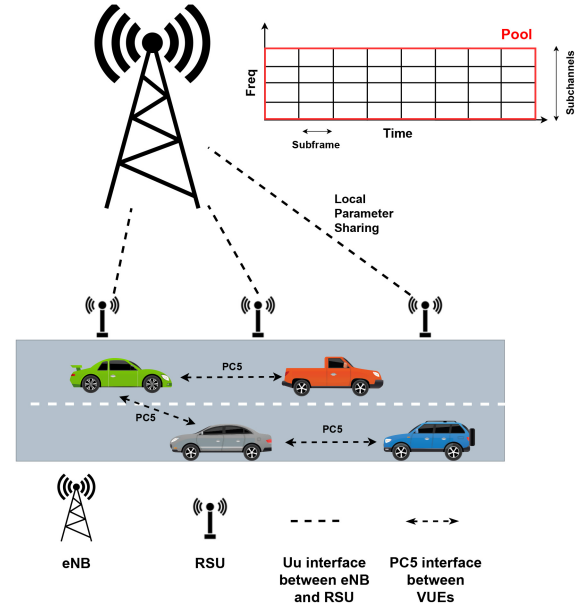


Fig. 1. System model.

To clarify the differences between our work and existing literature, Table I provides a comparison of the approaches, advantages, and limitations of the discussed studies.

IV. SYSTEM MODEL

The region is divided into cellular boundaries based on geographical coordinates. Each region is served by the base station, i.e., gNB. The Federal Communication Commission allocated the 75 MHz band at 5.9 GHz for V2X communication. From the 75 MHz, the spectrum is divided into regions and is preconfigured. In NR-V2X, vehicles communicate using resources that are distributed over the 2-D time and frequency domain. In the frequency domain, there are subchannels (spectrum bands) and in the time domain, there are subframes. The set of candidate subframes are represented as $C = \{c_1, c_2, c_3, \dots, c_K\}$. For instance, to represent a single candidate subframe resource, it is denoted as c_k . In each subframe there are number of subchannels. Each subframe consist of number of resource blocks. A single resource block is 360 kHz wide in the frequency domain with a 30 kHz subcarrier spacing. The gNB is preconfigured with the number of subchannels for the corresponding region. The set of gNB is represented as $g = \{g_1, g_2, g_3, \dots, g_N\}$. Each gNB has serving RSUs installed along the roadside, represented as $\eta = \{\eta_1, \eta_2, \dots, \eta_I\}$. η_i denotes the RSU where $i = \{1, 2, \dots, I\}$. η_I represent the number of RSUs connected to a single gNB. The gNB sends the configuration messages over SSB to the RSUs and the vehicles in its coverage range. Fig. 1 shows the system model. The RSU communicates with the vehicles in its coverage range via PC5 interface. The typical communication range of the PC5 interface is around 300 m, which allows RSUs to provide reliable connectivity within this range [40]. The set of vehicles is represented as $\phi = \{\phi_1, \phi_2, \dots, \phi_M\}$. ϕ_M represents the maximum number of vehicles that are in the coverage range of η_i . The vehicles from the group of

TABLE I
COMPARISON OF RELATED WORK

Reference	Approach	Advantages	Disadvantages
[24]	Congestion control with LBT and DCC	Improves congestion control in high channel load scenarios	Does not address decentralized resource scheduling in NR-V2X Mode 2
[25]	DCC with DRL	Enhances PDR and SLT through learning algorithms	Fails to address resource contention in NR-V2X Mode 2
[33]	Federated learning for V2V and V2I	Reduces interference between cellular users and VUEs	Does not focus on resource scheduling issues in NR-V2X Mode 2
[22]	Federated learning with mobility	Considers mobility in federated learning	Focuses on learning efficiency rather than resource contention
[31]	eSPS with reinforcement learning	Adapts resource reservations based on traffic patterns	It is not scalable. Does not consider distributive learning
[28]	Resource management for D2D	Optimizes D2D communication	Does not consider the NR-V2X Mode 2 standard
[17]	Federated learning for V2V and V2I communications	Maximizes sum rate of V2I communications	Does not address decentralized resource scheduling in NR-V2X Mode 2
[18]	Federated learning for mode selection in 5G-V2X	Optimizes mode selection between V2V and V2I	Lacks focus on resource scheduling required for NR-V2X Mode 2
[19], [20]	Federated learning for computational offloading and resource management in 6G-V2X	Optimizes vehicular edge computing	Does not address resource contention in NR-V2X Mode 2 communications
[21], [33]	Federated learning for vehicle selection and resource allocation	Enhances V2V communication and energy efficiency	Does not target decentralized resource scheduling in NR-V2X Mode 2
[22], [23]	Federated learning for joint communication, sensing, and computing resource allocation in CAV networks	Improves resource allocation in D2D-enabled 6G networks	Does not address specific challenges of NR-V2X Mode 2
[34]	Federated learning for SCMA in V2X	Enhances codeword design and reduces complexity	Does not address decentralized resource scheduling in NR-V2X Mode 2
[35]	Federated reinforcement learning for delay-sensitive IoV services	Maximizes spectrum-energy efficiency under delay constraints	Does not consider decentralized resource selection in NR-V2X Mode 2
[36]	Classical optimization vs machine learning in V2X networks	Compares resource allocation approaches for V2X platooning scenarios	Does not consider the NR-V2X standard and federated learning

allocated resources that are preconfigured utilize SPS to select the resources for their communication.

A. Semi-Persistent Scheduling

The vehicles select the resources from the selection window after sensing the resources from the sensing window. The size of the selection window is in the range of $n + (0, 4)$ ms to T_0 , where n shows the time at which the message packet arrives at the network layer from the application layer for transmission, and T_0 shows the packet delay budget. The sensing window is 1 s long, consisting of the last 1000 subframe resources before the message packet is generated, i.e., $(n - 1000)$ to $(n - 1)$. When a vehicle sends the message over the transport block utilizing the PSSCH, it also sends the corresponding control information by utilizing the physical sidelink control channel (PSCCH). Both channels consist of preconfigured resources. The PSCCH consists of resource reservation interval (RRI). The vehicle that is listening to the PSCCH perceives the RRI information that indicates that after that defined interval the channel is already reserved by that vehicle. Therefore, the vehicle performing the SPS restrains from selecting that resource.

Fig. 2 shows the SPS scheduling mechanism. The SPS consists of three steps.

- 1) Vehicles identify the list of available resources that it could reserve for the transmission of the transport block. It should be noted that the message packet is embedded over the transport block.
- 2) It excludes the resources and creates the list of candidate resources LA . The resources are excluded that meet the following two conditions simultaneously.
 - a) The resources that are already reserved based on the RRI indication.
 - b) The observed reference signal received power (RSRP) is higher than the predefined threshold.
- 3) From LA , vehicles create the list of candidate resources LC . LC must be at least $X\%$ (where $X = 20, 30$ or 50 depending on the traffic priority) of the resources that were identified in step 1. If not, the RSRP threshold is iteratively increased by 3 dB until the criteria is met.

From LC , the vehicle randomly selects the resources for its current transmission and reserves the resources for RC times with the interval of RRI. RC shows the retransmission counter which shows the number of remaining transmissions. With every consecutive transmission, RC is decremented by 1. When RC reaches zero, the vehicle broadcasts $RRI = 0$ to let the neighboring vehicles know that no more resource is occupied in the future by this vehicle. Also, for the next

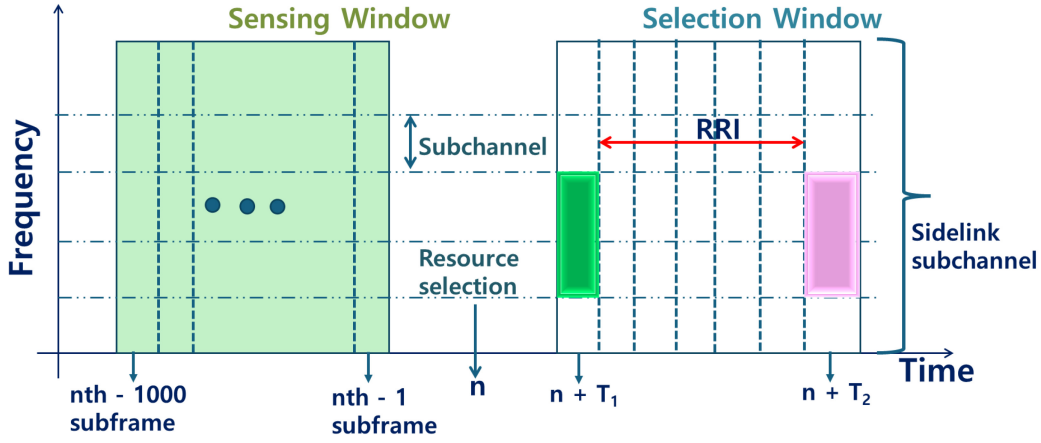


Fig. 2. SPS.

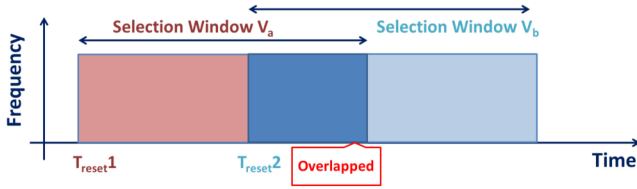


Fig. 3. Overlapped selection window.

time transmission, the vehicle performs the SPS again to select and reserve the resources for future transmission. The vehicle selects the resources with a $(1 - P)$ probability, where P is the resource keep probability and it varies between $(0.2, 0.8)$.

B. Problem Formulation

When a vehicle's sensing window overlaps with those of other vehicles, there is a higher probability that these vehicles will select the same resources, resulting in contention. Fig. 3 illustrates the overlapping sensing windows of two vehicles, v_a and v_b . Since v_a and v_b have overlapping sensing windows, due to the half-duplex nature, v_a and v_b are unaware of each other's resource selections. Therefore, there is a probability that both vehicles may select the same resources in step 3 of the SPS. When two vehicles select the same resource, it leads to interference, degrading the signal quality. A packet with a received signal quality above the outage probability is considered received; otherwise, it is discarded.

The probability of successful reception, conditioned on the SINR, is given by

$$\Pr\{\epsilon_{\phi_i} \geq \epsilon_0\} \quad (1)$$

where, ϵ_{ϕ_k} , is the SINR at vehicle ϕ_k , and ϵ_0 is the SINR threshold required for successful reception.

To model the data rate in this interference-limited environment, Shannon's capacity theorem is applied. The maximum achievable data rate u_k for a given SINR ϵ_k and channel bandwidth W is given as

$$u_k = W \log_2(1 + \epsilon_k) \quad (2)$$

where, u_k is the data rate achievable by vehicle ϕ_k , ϵ_k is the SINR for vehicle ϕ_k , and W represents the channel bandwidth.

The system's goal is to maximize the sum of the data rates across all vehicles, which can be expressed as

$$\max \sum_{k=1}^M W \log_2(1 + \epsilon_k). \quad (3)$$

This objective is subject to several constraints to ensure efficient communication. First, the SINR for each vehicle must exceed a required threshold ϵ_0 , ensuring reliable communication, is expressed as

$$\log_2(1 + \epsilon_k) \geq \log_2(1 + \epsilon_0) \quad \forall \phi_k \in \phi. \quad (4)$$

Equation (4) ensures that the data rate for each vehicle is above a minimum threshold based on the SINR requirement. Additionally, each vehicle's transmission power must be constrained within a maximum power limit $P_{\max}$, to avoid excessive interference. This constraint is expressed as

$$0 \leq P_k \leq P_{\max} \quad \forall \phi_k \in \phi. \quad (5)$$

Furthermore, each vehicle can select only one subframe from the set C at any given time, to prevent resource collisions. Let x_{ϕ_k, c_j} be a binary variable indicating whether vehicle ϕ_k selects subframe c_j ($x_{\phi_k, c_j} = 1$) or not ($x_{\phi_k, c_j} = 0$). The following constraint ensures that each vehicle selects exactly one subframe

$$\sum_{j=1}^K x_{\phi_k, c_j} = 1 \quad \forall \phi_k \in \phi. \quad (6)$$

Equation (6) guarantees that each vehicle selects one subframe at a time from the set of available resources C , although multiple vehicles may select the same subframe.

The overall goal is to maximize the achievable data rate while maintaining proper signal quality, power limits, and ensuring that each vehicle selects only one subframe from the available resources at any given time. The SINR ϵ_k for vehicle k is given in

$$\epsilon_i = \frac{P_k h_k}{\sigma + \sum_{j=1}^{N_j} x_{k,j} P_j h_j} \quad (7)$$

where, P_k is the transmission power of the vehicle, h_k is the channel gain between the transmitting and receiving vehicles,

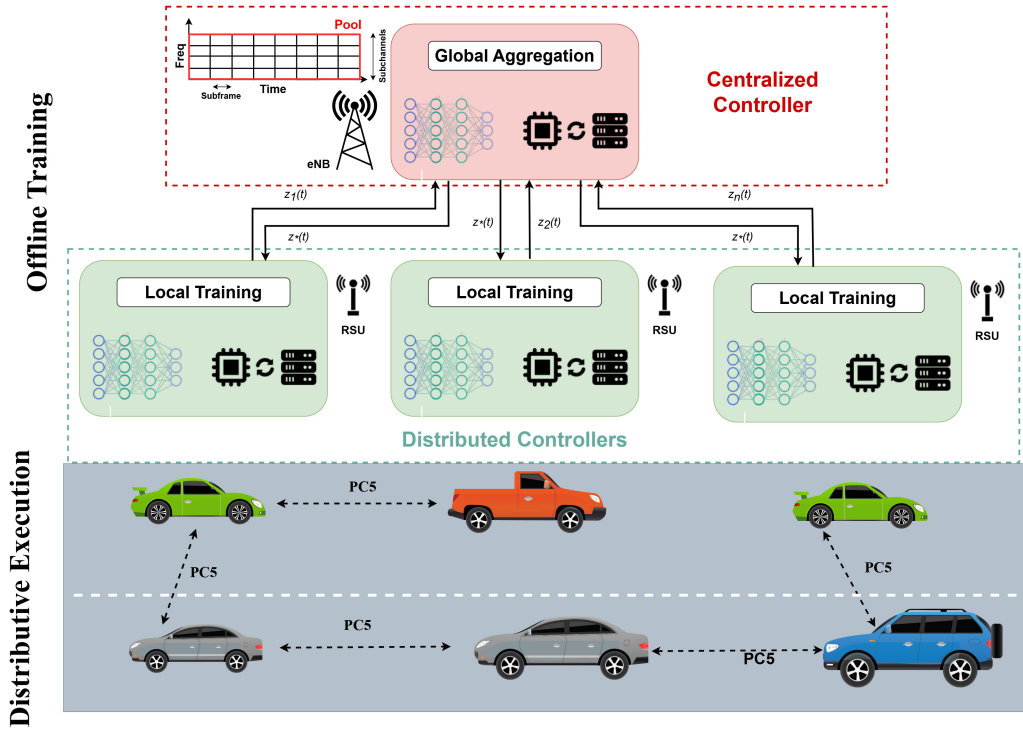


Fig. 4. Federated learning in V2X environment.

and σ represents the noise power. The binary variable $x_{k,j}$ denotes whether vehicles k and j use the same resource. P_j and h_j represent the interference power and channel gain from neighboring vehicles, and N_j is the number of neighboring vehicles contributing to interference.

The channel gain h_k is modeled as given

$$h_k = g_k \times \zeta \times A d^{-\gamma} \quad (8)$$

where, g_k accounts for small-scale fading, ζ represents the log-normal shadowing factor, A is the path loss constant, d is the distance between the transmitting and receiving vehicles, and γ is the path loss exponent.

The system also considers the packet reception ratio (PRR), which is defined as the probability that the SINR of a vehicle exceeds the threshold ϵ_0 , is expressed as

$$\text{PRR} = \frac{1}{N_j} \sum_{i=1}^{N_j} \Pr\{\epsilon_i \geq \epsilon_0\}. \quad (9)$$

In (9), N_j represents the number of neighboring vehicles. A packet is decoded successfully only if the received SINR exceeds the required threshold; otherwise, it is discarded to maintain the quality of communication, which is essential in V2V communication scenarios.

By maximizing the SINR for each vehicle, the data rate u_k is also maximized, as the achievable data rate is directly proportional to the SINR, as shown in (2). Therefore, efficient resource allocation and interference minimization will result in a higher data rate for each vehicle, ultimately improving PRR and overall network performance and ensuring reliable communication in vehicular networks.

V. PROPOSED FEDERATED LEARNING FRAMEWORK

Federated Learning is chosen as our machine learning approach to achieve the objective outlined in (3). This method offers several key advantages for our scenario. First, it allows us to train a global model while keeping sensitive data decentralized, addressing privacy concerns. This is particularly important in situations where data privacy and security are paramount. Additionally, federated learning promotes efficiency by minimizing the need for centralized data storage and transmission, which can be resource-intensive. Due to the mobility, having the vast region in a centralized manner is not efficient; therefore, we need a distributed approach. By harnessing the collective intelligence of local models on individual devices, we can collaboratively enhance the overall model's performance, leading to a maximized achievable rate for vehicles across the region. This distributed learning paradigm aligns with our objective of optimizing network performance while ensuring data privacy and operational efficiency.

The η within the coverage of g_i collects data samples $[\pi_{i1}, \pi_{i2}, \dots, \pi_{i\phi_m}]$ from ϕ for local training. The loss function at η_i is given as

$$L_i(Z_i) = \frac{1}{|S_i|} \sum_{\pi_i \in S_i} f(Z_i, \pi_i) \quad (10)$$

where, $|S_i|$ shows the size of the data sample, Z is the set of local parameters $\{z_1, z_2, \dots, z_n\}$, and $f(Z_i, \pi_i)$ represents the loss function for one data sample.

The federated learning works in rounds. Once the local parameters are obtained at the η , they are uploaded to the gNB for the global model. The local parameters are aggregated at the centralized gNB for the global model update. Fig. 4 shows

the federated learning in V2X environment. Once the global model is updated, the global parameters are downlinked to the η . The η updates the learning parameters and weights based on the global parameters. The model is iteratively updated until the desired accuracy is achieved. After completing all the local training, the local parameters obtained in the t th round are represented as $[Z_1(t), z_i(t), Z_n(t)]$, and the aggregated model of the global model is expressed as

$$Z(t) = \frac{1}{\sum_{i=1}^N |S_i|} \sum_{i=1}^N |S_i| Z_i(t). \quad (11)$$

A. Computation Model

The computation process involved in federated learning is the training of the local model [41]. The computation resources are defined as

$$r_\eta = \delta r \cdot y \quad (12)$$

where, δr represents the computation resource per unit and y represents the number of resources. The computation resources r_η are assigned to η for training on the local sample data of size $|S_i|$. The trained parameters Z are uploaded to the gNB via the dedicated cellular uplink.

Furthermore, the delay D_i incurred for updating the local model consists of the training delay T_D , uploading local parameters to the gNB delay $Upload_D$, downlink delay for the updated global parameters to η $Downlink_D$, and global model aggregation delay G_D . It is assumed that from the gNB to the RSUs there is enough bandwidth due to the dedicated downlink, therefore $Downlink_D$ is negligible. Similarly, the gNB, due to the central controller, has enough computing resources, therefore, G_D is also negligible and can be avoided. To this end, D_i is given as follows:

$$D_i = T_{D_i} + Upload_{D_i} = \frac{|S_i|}{r_\eta} + \frac{|Z_i|}{o_i} \quad (13)$$

where, o_i is the cellular uplink rate. For all participants η , the average delay D_{avg} is defined as follows:

$$D_{avg} = \frac{1}{N} \sum_{i=1}^N D_i. \quad (14)$$

B. Distributed Training With Centralized Topology

Distributed training is employed, where each RSU (η_i) collects local information from the vehicles (ϕ) in its vicinity to train a local model. Transmitting all data to a centralized server introduces significant communication overhead; therefore, federated learning reduces this overhead by only transmitting model updates, which are much smaller in size compared to raw data. The heterogeneity in data collected by different RSUs, such as varying traffic densities, vehicle speeds, and interference patterns, results in nonidentical and independent data (Non-IID) datasets. To address this, model aggregation at the gNB is performed using weighted averaging, where the contribution of each RSU is proportional to the number of vehicles in its coverage area. This ensures that

the global model captures the diversity of vehicular environments, enabling robust convergence. The updated global model weights are then sent back to each RSU (η_i) for dissemination to the vehicles, creating a feedback loop that improves the model iteratively.

For instance, consider two RSUs, RSU_A (η_1), and RSU_B (η_2). RSU_A covers an urban area with high vehicle density and significant resource contention, whereas RSU_B operates in a rural region with sparse traffic. RSU_A trains its local model with data reflecting high interference levels, while RSU_B optimizes for low-density traffic conditions. When their respective models are aggregated at the gNB, the global model combines these diverse datasets, ensuring that it can generalize across varying traffic scenarios. This process is iteratively refined as updated global policies are distributed back to the RSUs and vehicles, achieving convergence even with Non-IID data.

- 1) *State*: The RSU (η_i) collects sample data (π) from the vehicles (ϕ) in its coverage area. The state information used for local training is represented as

$$\eta_{(m,s_t)} = \{\phi_{C_1}, \phi_{C_2}, \dots, \phi_{C_M}\} \quad (15)$$

where, ϕ_{C_m} represents the measured RSRP and RRI over the subframe resources by vehicle ϕ_m . The complete state information at the vehicle level is given by

$$\phi_{(m,s_t)} = \{C_{RSRP_1, RRI_1}, C_{RSRP_2, RRI_2}, \dots, C_{RSRP_M, RRI_M}\} \quad (16)$$

where, $C = \{c_1, c_2, \dots, c_K\}$, shows the candidate subframe resources (subbands) from which the vehicle can select one for transmission.

In (16), C_{RSRP_M, RRI_M} represents both the RSRP and the RRI measured over all the subframe resources $C = \{c_1, c_2, \dots, c_K\}$ by vehicle M . The RSUs locally aggregate these state measurements from all vehicles in their coverage area. The state at the centralized gNB is then represented as

$$S_t = \{\eta_{(1,s_t)}, \eta_{(2,s_t)}, \dots, \eta_{(I,s_t)}\}. \quad (17)$$

It is worth noting that offline training is initially performed via simulation. Once the model is trained, the policies are sent to the RSUs (η) and vehicles (ϕ) over the SSB along with configuration messages. Following the principles of NR-V2X Mode 2, vehicles distributively select resources based on the trained policies received from the RSUs, avoiding resource contention.

- 2) *Action*: The action is to select one subband from $C = \{c_1, c_2, \dots, c_K\}$. Each vehicle is allowed to select only one resource from the list at a given time. To ensure this, the constraint is given in (6). The binary variable x_{ϕ_k, c_j} indicate whether the vehicle ϕ_k selects resource c_j ($x_{\phi_k, c_j} = 1$) or not c_j ($x_{\phi_k, c_j} = 0$). This ensures that each vehicle selects exactly one resource, preventing resource collisions.

- 3) *Reward*: The reward function is designed to maximize overall network performance, with the objective defined in (3). The reward is expressed as

$$R_t = \frac{1}{\phi_m} \sum_{\phi_m} \Pr\{\epsilon_{\phi_i} \geq \epsilon_0\}. \quad (18)$$

The agent utilizes a DQN during centralized training to select an optimal action based on the current state. As the agent interacts with the environment, it transitions to the next state s_{t+1} and receives the reward R_t , which reflects the effectiveness of its action. Through this iterative process, the agent learns and continuously refines the policy. Once the policy has converged to an optimal solution, it is transmitted to the vehicles for distributive execution. Importantly, the model is periodically retrained, with the training frequency adapting to changes in the environment, such as fluctuating traffic patterns or vehicular mobility.

The model's scalability and ability to handle Non-IID data are further enhanced through periodic retraining. As the number of vehicles or the complexity of the network increases, the system distributes the computational load across RSUs, preventing any single component from being overwhelmed. RSUs perform local training by collecting vehicular state information from their coverage areas and periodically sending model updates to the central controller. These updates are aggregated into a global model, which is then disseminated back to the RSUs and vehicles via the gNB. This global policy enables vehicles to make autonomous resource selection decisions, ensuring that the system remains scalable even in high-density networks.

To further manage scalability, the framework employs adaptive communication intervals, dynamically adjusting the frequency of global model updates based on network conditions. This reduces communication overhead in dense scenarios while maintaining model accuracy and performance. Additionally, a periodic learning mechanism at RSUs ensures that local models are retrained and updated as network conditions evolve. This iterative process allows the global model to remain effective, reflecting current traffic patterns, vehicle densities, and interference levels. The system also incorporates a local re-evaluation mechanism at each RSU, enabling vehicles to quickly adapt to local changes in their environment without waiting for continuous global model updates. This decentralized execution capability ensures robust and efficient performance, allowing the system to handle dynamic vehicular environments with varying traffic conditions and densities seamlessly.

VI. SIMULATION

We conducted simulations in a Manhattan grid scenario covering an area of $1250 \times 750 \text{ m}^2$, representing a dense urban environment with frequent intersections and traffic lights, which are typical of real-world urban vehicular networks. The vehicles were distributed across the scenario based on a Poisson distribution, emulating a realistic urban traffic pattern where vehicle density and traffic flow change dynamically. To simulate the movement and behavior of vehicles, we integrated

simulation of urban mobility (SUMO), a widely used traffic simulation tool, which enabled accurate modeling of vehicle mobility, speed variations, and traffic density in real time.

For this study, a single region was managed by a gNB, responsible for transmitting configuration messages and trained policies to RSUs within its vicinity. The gNB served as the central coordinator for resource allocation, training global models based on the aggregated local data collected by the RSUs. Four RSUs were strategically positioned along the roads within the grid, ensuring optimal coverage of the road network. These RSUs were equipped with mobile edge computing (MEC) capabilities, enabling them to perform local model training and aggregation in real time. MEC integration reduced latency in decision-making and resource allocation by allowing local computation at the RSUs, as opposed to relying solely on the gNB.

We considered a 10 MHz bandwidth, utilizing 25 resource blocks with a subcarrier spacing of 30 kHz, which aligns with the 5G NR specifications for V2X communications [7]. Each vehicle in the simulation acted as an independent agent within the federated learning framework, autonomously selecting resources based on trained policies received from the RSUs. This decentralized decision-making process mirrors real-world V2X systems, where vehicles select resources independently based on local observations rather than depending on a central controller.

The federated learning model was implemented through an iterative process to ensure convergence, even with Non-IID data collected from diverse vehicular environments. Initially, each RSU collected data from vehicles within its coverage area and trained local models based on this data. These local models were then transmitted to the gNB, where they were aggregated into a global model using weighted averaging. The weights for aggregation were proportional to the number of vehicles contributing data to each RSU, ensuring that regions with higher vehicle density had an appropriate influence on the global model. Once the global model was trained at the gNB, it was used to generate policies that were sent back to the vehicles over the SSB. This iterative process enabled the global model to capture the heterogeneity of the data and adapt to diverse traffic scenarios, ensuring robust convergence. The process mirrors how configuration parameters are sent to UE when forming the initial connection in cellular networks. The trained policies allow vehicles to make autonomous decisions on resource selection based solely on local observations, aligning with the NR-V2X Mode 2 resource selection process as specified by the 3GPP standard [8].

The rationale for sending trained policies to the vehicles is to ensure they have the necessary configuration parameters to make resource allocation decisions autonomously. For example, a vehicle operating in a high-traffic urban environment receives a policy optimized for minimizing resource contention, while a vehicle in a rural area applies a policy tailored for efficient resource usage in low-density conditions. Vehicles follow these policies without needing further input from the gNB, relying entirely on their local observations, such as vehicular state information and interference patterns. This enables vehicles to optimize resource selection in a

decentralized manner, minimizing the probability of resource collisions and improving overall network efficiency.

The distributed nature of this training process reduced communication overhead significantly, as only model updates, rather than raw data, were transmitted to the gNB. Moreover, by continuously refining the global model based on local data collected from diverse vehicular environments, the system adapts to changing traffic conditions, vehicle densities, and interference levels. For instance, RSU_A may send updates reflecting high resource contention due to urban congestion, while RSU_B provides updates optimized for sparse rural traffic. The gNB combines these updates into a unified global policy, ensuring that both scenarios are addressed effectively. This iterative refinement guarantees that the resource allocation policies remain accurate and up-to-date.

In real-world traffic conditions, the distributed local training and global model aggregation approach proves particularly effective in handling the dynamic nature of vehicular networks. The iterative training process accounts for the Non-IID nature of the data, as observations from various road conditions and traffic scenarios are incorporated into the global model. For example, vehicles in suburban areas with moderate traffic benefit from policies that balance resource contention and efficient usage, while vehicles in high-density city environments receive policies optimized for mitigating interference. These trained models enable efficient execution of resource allocation policies, reducing the likelihood of resource contention and improving overall system performance.

The proposed method's performance was compared to several approaches. First, we compared it to two related works [42] and [43]. Reference [42] uses a decentralized multiagent reinforcement learning scheme (multiagent DQN) for resource allocation in a heterogeneous V2X environment, and [43] addresses resource allocation challenges through Cluster-based scheduling (Cluster-based scheme). The comparison revealed that our approach enhances resource allocation efficiency and scalability, especially in dynamic vehicular environments.

In addition, we evaluated the results against a Greedy-based approach and random resource allocation. The Greedy-based method, which selects resources-based solely on immediate rewards, led to suboptimal performance, particularly under high traffic conditions, due to increased resource contention. The random allocation scheme, where resources are assigned randomly, further highlighted the superiority of our approach, as random selection resulted in significantly lower PRR and higher interference levels. Our method outperformed both in terms of PRR, system throughput, and adaptability to vehicular mobility.

Detailed simulation parameters are provided in Table II. Table II enumerates key variables and settings used in our experiments, offering a comprehensive overview of the simulation setup.

Fig. 5 illustrates the learning curve in our simulation, with a fixed learning rate of 0.0001, carefully selected to ensure smooth and stable training. The learning rate plays a vital role in balancing training speed and stability; an excessively large value could lead to erratic fluctuations in the training curve, while a smaller value might excessively slow down

TABLE II
SIMULATION PARAMETERS

Transmission power	23 dBm
No of subchannels	4
No of hidden layers	3
No of neurons	400
Channel	Winner + B1
Packet transmission rate	10 Hz
CAM size	{190, 300} bytes
Average vehicle speed	70 Kmhr^{-1}
No of RSUs	4
Computation resources per RSU	[5,30]
Learning rate	0.0001

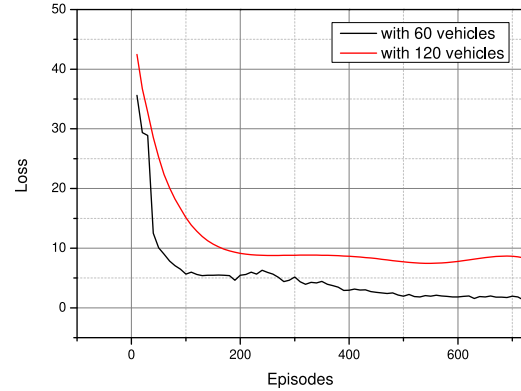


Fig. 5. Learning curve.

the convergence process. Therefore, the chosen learning rate ensures effective learning while avoiding sharp fluctuations.

The convergence plot in Fig. 5 shows the impact of varying the number of agents, with scenarios involving 60 and 120 agents. Notably, as the number of agents increases, convergence slows down. This deceleration is attributed to two primary factors: 1) the increased complexity of the environment with more agents and 2) the higher competition for limited communication resources. In large-scale vehicular networks, spectrum scarcity becomes a critical issue, leading to increased resource contention. This contention inevitably results in packet loss, despite the system's efforts to optimize resource allocation.

Fig. 5 demonstrate a general decrease in loss as episodes progress, indicating that the model is converging. However, slight fluctuations are observed during training. These fluctuations are inherent to the training process and reflect the model's dynamic adjustments as it learns to optimize resource allocation under varying network conditions. It is important to note that despite these fluctuations, the overall trend shows a consistent decrease in loss, confirming the model's convergence. Furthermore, a significant reduction in loss is observed after 200 Episodes, clearly indicating that the model is approaching its convergence point. The number of training episodes has been set to 700 to achieve a balance between training time and model performance. This choice ensures that the model has sufficient time to converge while maintaining computational efficiency.

Fig. 6 shows the average training delay, the training delay refers to the time it takes for RSU to compute local model updates

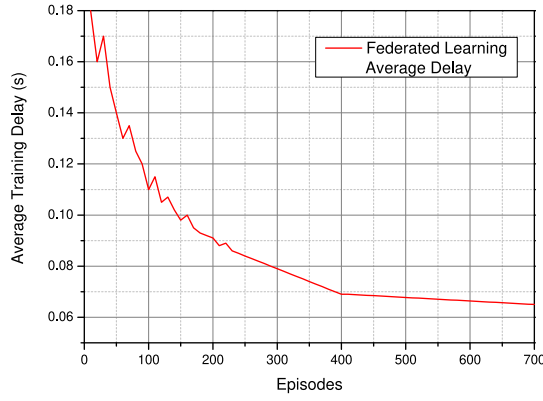


Fig. 6. Average training delay.

and send them back to the central server (gNB) for aggregation. This process is influenced by factors, such as network conditions, model complexity, and computational resources. Over time, the training delay decreases, as episodes progress, RSUs become more efficient in performing local computations, resulting in shorter training times. However, early in training, computations are intensive. As the model approaches convergence, the necessary computations become less demanding, leading to shorter training delays. This reduction in training delay reflects the improved efficiency of the federated learning process as the model progresses through the episodes.

For the performance evaluation, we compared the proposed federated learning-based deep reinforcement scheme with several approaches: the Greedy approach, DRL [42], Cluster-based approach [43], and Random resource selection. In the Greedy approach, vehicles consistently select channel indexes based on low observed reserved power, but this leads to a high probability of multiple vehicles selecting the same channel, causing significant resource contention. The Cluster-based approach, which groups vehicles based on geographic regions and reuses channels among clusters, performs better with fewer vehicles but struggles as vehicle density increases due to inefficient resource allocation. The DRL method, which trains policies centrally using a single agent, also faces performance degradation in high mobility scenarios as vehicle density rises, as its centralized nature limits adaptability to dynamic traffic conditions.

In contrast, our proposed federated learning-based method excels due to its distributed training architecture, where RSUs locally collect and process data based on real-time traffic conditions. These local updates are then aggregated at the gNB, ensuring that the global model reflects the dynamic and heterogeneous vehicular environment. This distributed training approach enables the local models to capture the unique characteristics of their immediate environments. Once aggregated into a global model, the learned policies are more robust and adaptable to the full spectrum of vehicular scenarios. Consequently, vehicles can make efficient resource allocation decisions in real-time, minimizing resource contention and improving overall network performance.

To evaluate the effectiveness of our approach, we conducted experiments under a scenario where vehicles generate packets

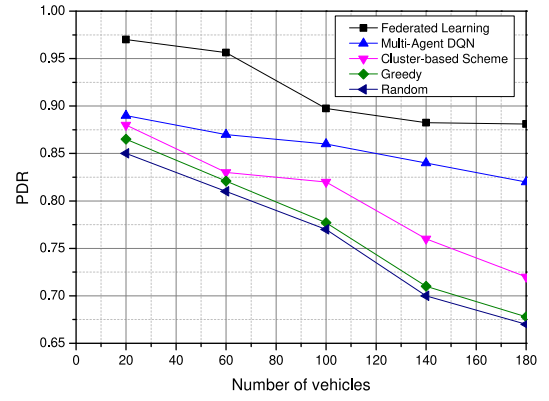


Fig. 7. PDR with respect to number of vehicles.

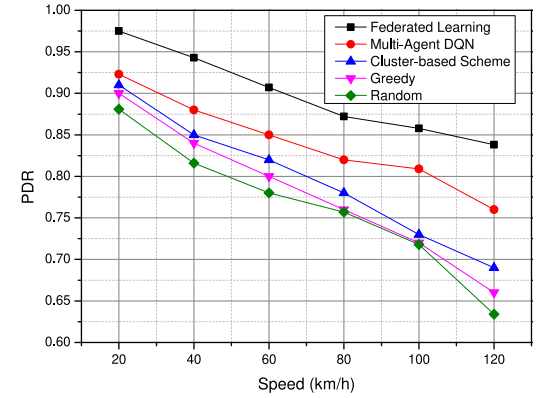


Fig. 8. PDR with respect to speed.

at a 10 Hz rate with 200 Bytes payload. Despite inevitable spectrum scarcity and increased resource contention as the number of vehicles increases, our federated learning method consistently outperforms other approaches in terms of PDR and scalability. As illustrated in Fig. 7, the federated learning framework excels in adapting to localized traffic conditions through distributive training at RSUs, allowing for better scalability and resource allocation efficiency. This advantage is especially evident in dense traffic scenarios, where the decentralized nature of federated learning leads to reduced contention and superior PDR compared to centralized DRL and other methods. The ability to handle spectrum scarcity and resource competition effectively demonstrates the robustness of our approach in real-world high-density vehicular environments.

In Fig. 8, with the increase in vehicular speed, a consistent trend of performance degradation is observed in all cases, primarily due to the heightened dynamics and challenges associated with faster vehicle movements. However, it is worth noting that the performance degradation is notably mitigated when utilizing a federated learning-based approach. The distributed nature of federated learning facilitates more adaptive and efficient updates in response to the increased speed of vehicles, resulting in a relatively superior performance when compared to nonfederated methods. This demonstrates the resilience and effectiveness of the federated learning approach in handling the challenges posed by higher vehicular speeds.

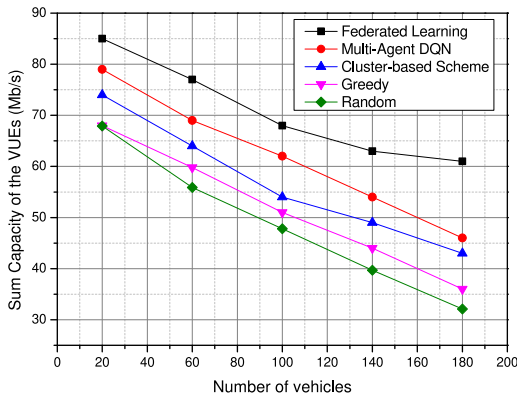


Fig. 9. Average rate of the VUEs with respect to number of vehicles.

Fig. 9 illustrates the average data rate of VUEs as the number of vehicles increases, with a fixed packet size of 180 bytes. As the number of vehicles rises, the graph reveals a consistent decline in the average data rate. This diminishing trend is directly linked to the degradation in the SINR, which worsens due to the increased interference from neighboring vehicles competing for the same limited resources. Spectrum scarcity exacerbates this situation, as more vehicles contend for the available communication channels.

Despite these challenges, the federated learning-based approach demonstrates a significant performance advantage compared to other methods, such as DRL, Greedy-based, Cluster-based, and Random resource allocation schemes. The fixed packet size of 180 bytes was selected to emulate typical safety message sizes in vehicular communication, ensuring a realistic evaluation of performance. Even with this fixed packet size, the federated learning approach excels by leveraging locally trained models with global updates. This mechanism enables vehicles to reuse resources strategically when they are distant from one another, while minimizing interference in close proximity. As a result, the overall network performance is enhanced, as vehicles can make more efficient resource allocation decisions in real-time.

In contrast, DRL without federated learning shows limited scalability and adaptability, particularly as vehicle density increases, due to its centralized nature. Similarly, the Greedy and Cluster-based approaches suffer from higher resource contention and inefficient allocation strategies, leading to lower average rates as vehicle numbers grow. In conclusion, while spectrum scarcity and increased vehicular density naturally lead to performance degradation, the federated learning-based approach mitigates these effects better than the alternative methods, maintaining higher data rates and more efficient resource utilization.

Fig. 10 presents the average data rate of VUEs with varying packet sizes, keeping the number of vehicles fixed at 40. As the packet size increases, there is a noticeable decline in the average data rate. This phenomenon occurs because larger packet sizes require more channel resources for transmission, increasing contention for the same set of resources. The decline in average rate due to larger packet sizes reflects a similar effect seen when more vehicles contend for limited resources.

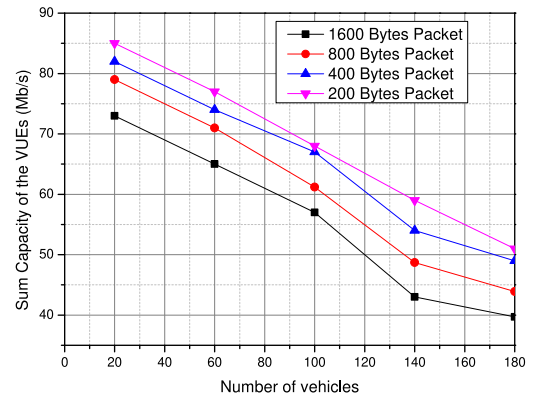


Fig. 10. Average rate of the VUEs with respect to varying packet size.

Despite this inevitable decline in performance caused by spectrum scarcity, the federated learning approach remains effective in optimizing resource allocation, even when packet sizes vary. The decentralized nature of the federated learning framework allows for adaptable and efficient adjustments to changing packet size requirements, ensuring better performance compared to other schemes. The ability to dynamically distribute learning across RSUs and aggregate local model updates at the gNB enables the system to handle varying traffic conditions and packet sizes more effectively. In contrast, alternative methods, such as Greedy-based, Cluster-based, and Random allocation struggle to maintain efficient resource utilization when packet sizes increase, highlighting the federated learning approach's superiority in enhancing overall vehicular communication system performance.

Fig. 11 illustrates the performance of the proposed scheme under different load scenarios. The number of vehicles varies, and the evaluation is conducted with packet generation rate, such as 10 Hz and 20 Hz. Additionally, packets of varying sizes 200 Bytes, 800 Bytes, and 1600 Bytes are considered. It has been observed that the proposed scheme performs well even in high-density scenarios, particularly with 180 vehicles, a message generation interval of 20 Hz, and a packet size of 1600 Bytes. Fig. 7 presents a comparison of the proposed scheme with different methods, including multiagent DQN, Cluster-based schemes, the Greedy method, and the Random method. From Fig. 11, it is evident that even at a 10 Hz packet generation rate and a packet size of 200 Bytes, the PDR drops to approximately 0.82 in multiagent DQN, whereas in the proposed scheme, the PDR remains above 0.88. Similarly, from Fig. 11, it can be observed that with a packet generation rate of 20 Hz and a packet size of 800 Bytes, the PDR drops to 0.85 with 180 vehicles. However, the performance remains satisfactory. Therefore, the proposed scheme proves to be effective even under high load conditions.

In Fig. 12, the convergence of training is depicted for both federated learning-based DRL and traditional DRL without federated learning. The graph illustrates that federated learning consistently yields higher rewards, indicating lower training loss when compared to the conventional DRL approach. Notably, the training stability is achieved more swiftly in federated learning, as evidenced by the convergence with fewer episodes when compared to the DRL approach. This implies

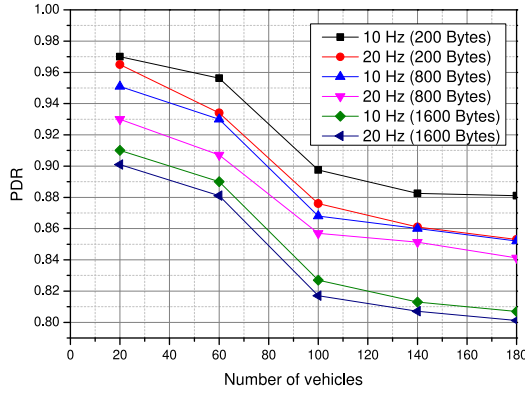


Fig. 11. Proposed scheme evaluation with varying packet size and generation rate.

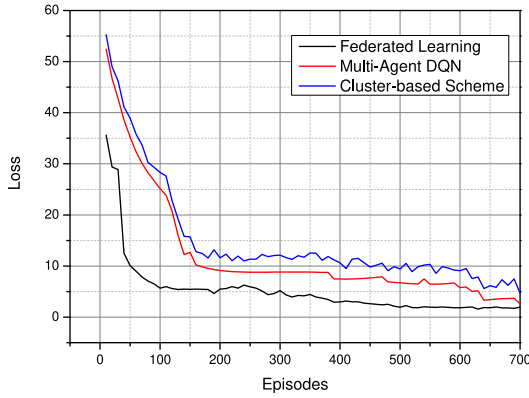


Fig. 12. Training convergence.

that the federated learning methodology not only enhances performance in terms of training efficiency but also achieves a more stable and reliable convergence during the learning process.

VII. COMPLEXITY ANALYSIS

This section presents the complexity analysis of the proposed federated learning framework for NR-V2X Mode 2 resource scheduling and compares it with traditional centralized training approaches. Although both methods share some common computational terms, federated learning offers significant practical advantages in terms of communication efficiency, parallel computation, and overall scalability in large vehicular networks.

In our proposed approach, each vehicle monitors K subbands and records its RSRP and RRI. For N vehicles, the total state space complexity is proportional to the number of vehicles and subbands, which can be expressed as $O(N \times K)$. Both centralized and federated learning approaches must process this state information to train a model for resource allocation.

In the traditional centralized training method, each vehicle transmits its full state information (RSRP, RRI) to a central controller, such as a gNB, where the entire model is trained. This approach introduces significant communication overhead because the raw state data from all vehicles must be sent to the gNB. The complexity of communication in this centralized

scenario is $O(N \times K)$, representing the cost of transmitting the state information of N vehicles across K subbands.

Once the data is collected at the gNB, the model is trained using optimization techniques, such as stochastic gradient descent. The computational complexity of the centralized training process, which scales with the number of vehicles N , the number of subbands K , the number of model parameters P , and the number of gradient updates G , is given by

$$C_{\text{central}} = O(R \times P \times N \times K \times G) \quad (19)$$

where, R represents the number of training rounds. Therefore, the total time complexity of centralized training can be written as

$$O(T_{\text{central}}) = O(R \times P \times N \times K \times G) + O(N \times K). \quad (20)$$

In (20), the first term corresponds to the training complexity, and the second term represents the communication overhead. In large networks, the communication cost of transmitting raw data can become a major bottleneck.

In contrast, the proposed framework distributes the computational load across multiple RSUs. Each RSU performs local training based on the data from the vehicles within its coverage area. This reduces the communication overhead and allows the training process to occur in parallel across different RSUs.

The complexity of local training at each RSU η_i , which processes the data from n_i vehicles, is given by

$$O(\mathcal{L}_{\eta_i}) = O(R \times P \times n_i \times K \times G) \quad (21)$$

where, n_i represents the number of vehicles under RSU η_i , and G is the number of gradient updates.

The total complexity for local training across all RSUs is

$$O(\mathcal{L}) = O\left(R \times P \times \sum_{i=1}^I (n_i \times K \times G)\right) \quad (22)$$

since $\sum_{i=1}^I n_i = N$, the overall complexity for local training simplifies to

$$O(\mathcal{L}) = O(R \times P \times N \times K \times G). \quad (23)$$

After local training, the RSUs send the learned model parameters to the central controller (gNB) for aggregation. The aggregation process, which combines the local models from all RSUs, has a complexity of $O(R \times P \times I)$, where I represents the number of RSUs. Since the number of RSUs I is typically much smaller than the number of vehicles N , this aggregation step adds only a minimal overhead.

The communication overhead in federated learning is significantly lower compared to centralized training because each RSU transmits model updates rather than raw data. The complexity of this communication step is:

$$O(\mathcal{H}) = O(P \times I). \quad (24)$$

Thus, the total time complexity of the federated learning approach is

$$O(T_{\text{total}}) = O(R \times P \times N \times K \times G) + O(R \times P \times I) + O(P \times I). \quad (25)$$

In (25), $O(R \times P \times N \times K \times G)$ represents the local training that happens in parallel at the RSUs, $O(R \times P \times I)$ accounts for the model aggregation at the gNB, and $O(P \times I)$ represents the communication cost of transmitting the model updates from RSUs to the gNB.

A. Comparative Analysis and Assumptions

Both the centralized and federated learning approaches share the same core training complexity term $O(R \times P \times N \times K \times G)$, which accounts for the time spent training the model. However, the practical benefits of federated learning emerge from two key aspects.

- 1) *Parallel Local Training*: In federated learning, the local training is distributed across multiple RSUs, allowing the training to happen in parallel. This parallelism significantly reduces the time required for training, as the computational load is spread across several RSUs rather than being concentrated at a single gNB, as in centralized training.
- 2) *Reduced Communication Overhead*: In centralized training, the communication cost of transmitting raw state data from N vehicles to the gNB is $O(N \times K)$, which can dominate the overall complexity in large networks. In contrast, federated learning only requires the transmission of model updates, with a communication complexity of $O(P \times I)$, which is much smaller and more manageable in practice.

Therefore, while both approaches may appear to have similar computational complexity in terms of Big-O notation, the parallelism and communication efficiency of federated learning make it far more scalable and suitable for large, dynamic vehicular networks. In particular, the communication term $O(N \times K)$ in centralized training can become a significant bottleneck, whereas federated learning avoids this by only transmitting model updates, reducing the communication complexity to $O(P \times I)$. Hence, federated learning offers substantial improvements in both communication efficiency and time complexity by distributing the training process and reducing the need for transmitting raw data. This makes proposed scheme more scalable, efficient, and better suited for dynamic environments, such as NR-V2X Mode 2.

VIII. CONCLUSION

In this article, federated learning is proposed for the distributive resource selection in NR-V2X Mode 2. The training process is carried out distributively via an offline method, specifically at the RSUs positioned along the road. This distributive training has shown commendable accuracy in the conducted performance evaluations, primarily attributed to the locally gathered environmental sample dataset at the RSUs. The model undergoes iterative training, with weights being updated based on the global model until the desired level of accuracy is achieved. Subsequently, the trained model policies are disseminated to the vehicles through RSUs over the PC5 interface, enabling them to independently select resources in real-world vehicular scenarios. Based on the implicit relation between the actions rewards and the states, vehicles select

the resources to increase the overall network performance by mitigating the resource contention.

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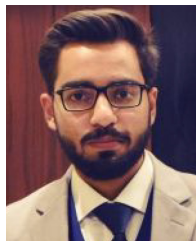
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